# Sahaj Bhadja

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## Summary

Software engineer pursuing an MSCS at USC with 2+ years building data-heavy software systems, SQL-backed workflows, and tested service integrations. Strong mathematical CS foundation, production debugging experience, and interest in ramping quickly into APL/Dyalog for wealth platform delivery.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

## Experience

**USC Dornsife** **Los Angeles, CA**
*Software Engineer* *September 2025 – Present*
- Designed and optimized SQL-backed workflows supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and service-level fixes.
- Built validation pipelines processing 5,000+ monthly jobs with reconciliation checks, CI/CD releases, and clear runbooks that improved triage speed for data and integration failures.

**Grad8** **Los Angeles, CA**
*Software Engineer Intern* *June 2025 – August 2025*
- Developed React and Flask services handling 10,000+ daily requests, implementing REST APIs, debugging database states, and writing tests that improved release confidence.
- Built batch-processing workflows for PDF, DOCX, and ZIP inputs with AWS-backed persistence, reducing feedback turnaround by **60%** while documenting edge cases and failures.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*
- Built Python evaluation pipelines across 100,000+ records, producing reproducible reports, debugging artifacts, and statistical comparisons that clarified model behavior and data quality.
- Analyzed failure patterns across generated outputs and datasets, translating technical findings into concise recommendations while strengthening validation coverage and engineering handoffs.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and accelerating defect isolation across services.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular services and responsive interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

## Skills

**Languages**: Python, Java, TypeScript, JavaScript, SQL, C++, Bash, APL/Dyalog ramp-up interest
**Data & Files**: PostgreSQL, MySQL, SQL Server, MongoDB, CSV/flat-file processing, XML familiarity, data validation
**Engineering**: REST APIs, microservices, unit testing, integration testing, debugging, technical documentation, production support
**Delivery**: Git/GitHub, version control, Docker, AWS, Jenkins, CI/CD, code reviews, requirements analysis, Agile collaboration

## Projects

**Chat2SQL** Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, combining schema inspection, query generation, Dockerized services, and debugging flows for faster data validation.

**FinNexus** Code | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a secure fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, scalable data handling, and service-backed financial workflows.

**CPU Scheduling Simulator** Live | *C++, React.js, JavaScript*
- Implemented scheduling algorithms and visual comparisons for operating-system workloads, reinforcing mathematical reasoning, data structures, algorithm analysis, and clear explanation of system behavior.